

Supplementary diagnostics and implementation figures

Crowd Supervision and Vocabulary-Mismatched Selective Facial-Expression Prediction

SN Computer Science

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Interpretation and evidence boundaries

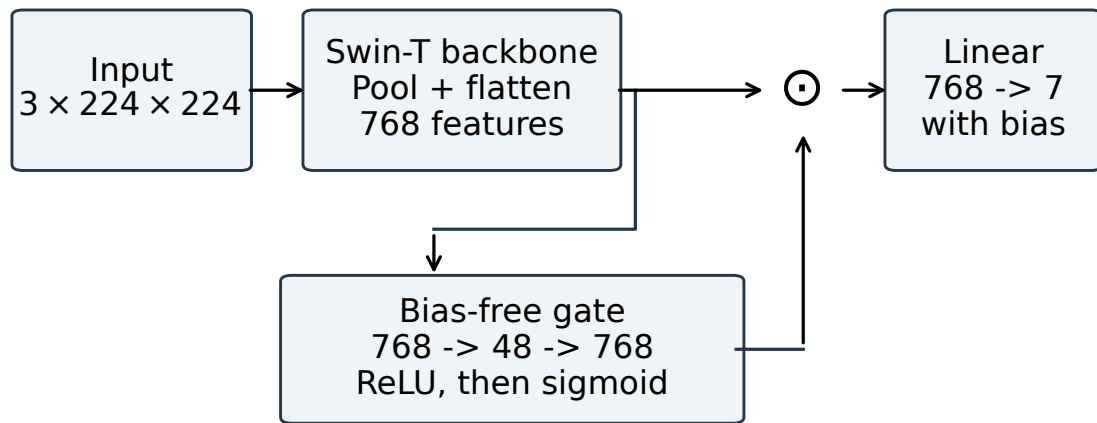
All numerical summaries are generated from `reported_results.csv`. S denotes Soft, U Uniform, H Hard, and T Tie-hard. Differences are percentage points. Three seeds are fixed, not samples supporting broad seed-population inference. The image source is reused; no external-domain or communication benefit is established. Global 80% refers to restricting a global accepted set to a stratum, whereas within-stratum 80% reranks that stratum independently. Supported-only risk conditions the vote reference on supported categories and is undefined at zero supported mass.

Table S1: Every paired seed contrast at 80% coverage

State	Contrast	Seed 17	Seed 42	Seed 89
selected	Soft - Uniform	-0.3473	-0.4313	-0.3550
selected	Soft - Hard	-2.1908	-3.7328	-1.9084
selected	Tie-hard - Hard	-0.0153	-1.0992	-0.2977
selected	Soft - Tie-hard	-2.1756	-2.6336	-1.6107
epoch12	Soft - Uniform	-0.3282	-0.3931	-0.2290
epoch12	Soft - Hard	-1.1794	-1.4198	-1.4542
epoch12	Tie-hard - Hard	-0.2824	-0.5076	-0.4313
epoch12	Soft - Tie-hard	-0.8969	-0.9122	-1.0229

Table S2: Selected epochs from the common squared-distance criterion. Every fixed-epoch sensitivity checkpoint is epoch 12

Targets	Seed 17	Seed 42	Seed 89
Hard	3	2	5
Tie-hard	4	4	5
Uniform	9	6	10
Soft	9	10	10



One gate after pooling; 73,728 additional parameters

Figure S1: Fixed model head used in all four conditions: one channel gate after pooled Swin-T features, followed by a seven-category linear classifier. This is not a new architecture claim

Local expression-cue prototype: illustrative controls

Synthetic text and score; not a participant record or measured prediction

Local camera preview No facial image shown Example: neutral Score: 0.62 (uncalibrated)	Compose a message I will send the draft tomorrow. ☐ Attach current expression cue Add to local preview
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Default: off; camera off or invalid/stale observation: omit the cue

Opt-in resets after each message. No face does not mean neutral.

Messages remain local; no recipient-side transport is implemented.

Figure S2: Synthetic illustration of the local sender-controlled interface. Text and category presentation are illustrative, not participant observations or a measured prediction. No recipient transport is implemented

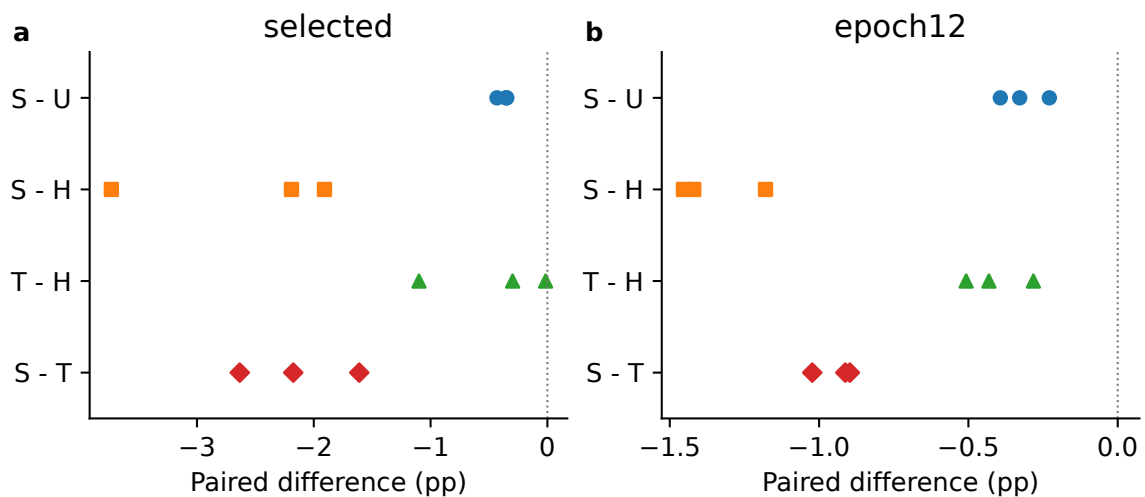


Figure S3: Individual paired seed differences at the common-coverage endpoint: (a) selected and (b) fixed epoch-12 weights. Exact values appear in Table S1; points can overlap

Table S3: Supported-mass strata: means over three selected-checkpoint seeds. Risk columns are percentages; accepted count can vary by seed

Targets	Stratum	Selection	N	Accepted	Risk	Supported-only
Hard	full	all	1732	1732.00	19.93	19.93
Hard	full	global80	1732	1497.33	14.96	14.96
Hard	full	within80	1732	1386.00	12.80	12.80
Hard	partial	all	1527	1527.00	47.53	38.44
Hard	partial	global80	1527	1116.33	41.08	31.30
Hard	partial	within80	1527	1222.00	42.69	33.04
Hard	zero	all	16	16.00	100.00	–
Hard	zero	global80	16	6.33	100.00	–
Hard	zero	within80	16	13.00	100.00	–
Tie-hard	full	all	1732	1732.00	19.42	19.42
Tie-hard	full	global80	1732	1507.00	14.51	14.51
Tie-hard	full	within80	1732	1386.00	12.33	12.33
Tie-hard	partial	all	1527	1527.00	46.84	37.52
Tie-hard	partial	global80	1527	1105.67	40.76	30.74
Tie-hard	partial	within80	1527	1222.00	42.23	32.38
Tie-hard	zero	all	16	16.00	100.00	–
Tie-hard	zero	global80	16	7.33	100.00	–
Tie-hard	zero	within80	16	13.00	100.00	–
Uniform	full	all	1732	1732.00	18.32	18.32
Uniform	full	global80	1732	1523.00	13.45	13.45
Uniform	full	within80	1732	1386.00	10.94	10.94
Uniform	partial	all	1527	1527.00	45.71	36.24
Uniform	partial	global80	1527	1090.00	38.40	28.24
Uniform	partial	within80	1527	1222.00	40.58	30.62
Uniform	zero	all	16	16.00	100.00	–
Uniform	zero	global80	16	7.00	100.00	–
Uniform	zero	within80	16	13.00	100.00	–
Soft	full	all	1732	1732.00	17.93	17.93
Soft	full	global80	1732	1529.00	13.23	13.23
Soft	full	within80	1732	1386.00	10.42	10.42
Soft	partial	all	1527	1527.00	45.36	35.75
Soft	partial	global80	1527	1080.00	37.71	27.35
Soft	partial	within80	1527	1222.00	40.00	29.77
Soft	zero	all	16	16.00	100.00	–
Soft	zero	global80	16	11.00	100.00	–
Soft	zero	within80	16	13.00	100.00	–

Full means all votes are supported; partial means positive supported and unsupported mass; zero means no supported votes. A risk of one in the zero stratum follows from the vocabulary, not a special model failure. Global selection may leave some strata empty; risk then remains undefined.

Table S4: Calibration half-split holdout-minus-fit risk gaps. Means and 5th/95th percentiles are descriptive percentage points over nonempty seed-split outcomes, not confidence intervals

Targets	Target	Nonempty	Mean gap	5th pct.	95th pct.
Hard	10%	300/300	-0.040	-2.215	+2.175
Hard	20%	300/300	+0.050	-2.328	+2.590
Hard	30%	300/300	+0.008	-2.752	+2.598
Tie-hard	10%	300/300	-0.002	-2.169	+2.399
Tie-hard	20%	300/300	+0.180	-2.344	+2.770
Tie-hard	30%	300/300	+0.072	-2.683	+2.859
Uniform	10%	300/300	+0.216	-1.808	+2.033
Uniform	20%	300/300	+0.193	-2.021	+2.617
Uniform	30%	300/300	+0.037	-2.442	+2.536
Soft	10%	300/300	+0.060	-1.871	+1.736
Soft	20%	300/300	+0.170	-2.004	+2.428
Soft	30%	300/300	+0.015	-2.410	+2.736

Each of 100 shared pixel-cluster splits uses disjoint calibration halves for fitting and checking the empirical threshold. Splits overlap one another. Minimum fitted accepted count is 100 and score ties are never split. The full ledger includes fit/holdout counts, risks, thresholds, and holdout coverage for every split, including empty policies.

Unconditional population composition

Calibration: unsupported mass 5.7247%, supported ambiguity 17.9720%, annotation floor 23.6967%.

Test: unsupported mass 7.8779%, supported ambiguity 18.7878%, annotation floor 26.6656%.

These annotation-only quantities do not depend on the fitted model. Accepted-set differences in the main figure condition on each model’s chosen threshold and can differ from these unconditional differences. None uniquely identifies a causal source of overshoot.

Reproduction map

REPRODUCIBILITY.md provides CPU numerical-audit and optional image-dependent inference commands. Online Resource 1 includes predictions, source hashes, category maps, original-row joins, both protocols and tests. Online Resources 2–5 contain all selected and epoch-12 inference weights. No face images are redistributed. Internal automated verification is not external replication.